

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR (COMPUTER SCIENCE & INFORMATION TECHNOLOGY/ COMPUTER SCIENCE
SPECIALIZATION IN "ARTIFICIAL INTELLIGENCE")
FALL SEMESTER EXAMINATIONS 2024
Batch - 2022

Time: 3 Hours

Dr. Maria Andleeb

Dated: 13-12-2024

Max. Marks: 60

OPERATING SYSTEMS - CT-353

Question 1

[C4 – 12 marks]

- a) Analyze the following set of processes, with the length of the CPU burst given in milliseconds. (6)

Process	Burst Time	Priority
P ₁	10	3
P ₂	1	1
P ₃	2	3
P ₄	1	4
P ₅	5	2

The processes are assumed to have arrived in the order P₁, P₂, P₃, P₄, and P₅ all at time 0.

- Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1).
 - What is the average waiting time and average turnaround time of each process for each of the scheduling algorithms in part i?
- b) Compare different multithreaded models. (3)
- c) Linked the use of Light weight process (LWP) in scheduler activations. (3)

Question 2

[C4 – 12 marks]

- Illustrate the windows threads data structures diagrammatically. (4)
- Explain the thread cancellation in detail. (4)
- Discuss Amdahl's Law and compare concurrency and parallelism. (4)

Question 3

[C4 – 12 marks]

- a) Analyze the system with 5 processes <P₀, P₁, P₂, P₃, P₄> and three resources types A=10, B=5 and C=7. The following information has been given. (6)

Allocation Matrix				Max Matrix			
	A	B	C		A	B	C
P ₀	0	1	0	P ₀	7	5	3
P ₁	2	0	0	P ₁	3	2	2
P ₂	3	0	2	P ₂	9	0	2
P ₃	2	1	1	P ₃	2	2	2
P ₄	0	0	2	P ₄	4	3	3

Available resources are A=3, B=3, C=2.

- Find need Matrix
- Is the system in safe state?? (If yes find the safe sequence using Banker's Algorithm).

- b) Elaborate the scenario of RPC communication. (3)
- c) Discuss semaphore implementation with no busy waiting. (3)

[C2 – 12 marks]

Question 4

- Give reason why monitors are not powerful enough to model some synchronization schemes. (3)
- Define the terms:
 - Deadlock
 - Starvation
 - Priority Inversion
- Explain the dining philosopher's problem. (3)
- Differentiate between preemptive and non-preemptive approaches of critical section handling in OS. (3)

Question 5

[C2 – 12 marks]

- List the advantages of multi-threaded system architecture. (3)
- Discuss process states with the help of diagram. (3)
- Differentiate between synchronous and asynchronous inter process communication of message passing. (3)
- Discuss the components of Process Control Block. (3)

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR(Bachelor of Science In Computer Science & Information Technology)
FALL SEMESTER EXAMINATIONS 2023

Batch 2021

Dated : 31-JAN-24

Max Marks : 60

Time : 3 Hours

Dr NajeedOperating Systems - CT-353

Instructions: Attempt all questions. All questions carry equal marks.

✓ Q.No 1. Define the following terms in operation system: Logical and Physical address, Virtual and Cache memory, Process and Threads. CLO-1

✓ Draw block only diagram and label them to Interpret Long-term scheduler, Mid-term scheduler and Short-term scheduler in Operating System. CLO-1

Q.No 2. What is computer Bus, What are the bus design issues? State the difference between PCI bus, ISA bus and SCSI bus. CLO-3

✓ Where Feedback Multi Level Queue algorithm of scheduling is used? How it is effective as compared to other algorithms. CLO-3

✓ Q.No 3. What is the difference between Blocked state and Deadlock state process? Also state the Deadlock prevention and avoidance scheme. CLO-2

✓ Q.No 4. What are the placement algorithms, write the names of placement algorithms used for memory allocation. Define Next Fit algorithm in detail. CLO-2

✓ Q.No 5. Following set of processes A, B, C, D, E with the length of the CPU burst time, Priority (high to high) and arrival time are given in table: CLO-1

Process	Burst Time millisecond	Priority	Arrival Time
A	5	2	1
B	1	1	0
C	6	3	1
D	10	4	5
E	4	3	3

Find the average waiting time, and draw Gantt chart for:

- (a) FCFS Scheduling (preemptive)
 (b) Round robin with quantum time $q = 4$ millisecond.

Dated: 11-02-2023
Max.Marks:60

Time: 3 Hours

Operating Systems - CT-353

Instructions: Attempt all questions. All questions carry equal marks.

Q.No 1 Define the following terms in the field of operation system: Burst, Time, Quantum time, Seek time, Response time, Waiting time, Turnaround time.
Draw block diagram only to show Long-term scheduler, Mid-term scheduler and Short-term scheduler placement between the storage devices, Processor and I/O devices.

Q.No 2 (a). What are the different kinds of Interrupts in operating system, define each with example? How an Interrupt can be handled?

(b) Where Multi Level Queue algorithm of scheduling is used? How it is effective as compared to other algorithms.

Q.No 3 What is the difference between Blocked state and Deadlock state process? Also state the Deadlock avoidance scheme.

(b) Compare the Preemptive and Non-preemptive OS scheduling modes?

Q.No 4. (a) What are the placement algorithms, write the names of placement algorithms used for memory allocation. Define Next Fit algorithm.

(b) Define pages and frames?

Q.No 5. Following set of processes A, B, C, D, E with the length of the CPU burst time, Priority (high to high) and arrival time are given in table:

Process	Burst Time	Priority	Arrival Time
A	5	3	1
B	1	1	0
C	6	3	1
D	10	4	6
E	4	2	4

Find the average waiting time, and draw Gantt chart for:

(a) SJFS Scheduling (preemptive)

(b) Round robin with quantum time $q = 3$ millisecond.

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DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
TSCS, Midterm Examinations Fall 2022

Time: 90 minutes

Operating System (CT-353) - A

Max Marks: 20

Note: Attempt all questions.

✓ Q-1) What is an Operating system responsible for? Describe the basic Components of an Operating system? (5)

✓ Q-2) Why the placement algorithms use in OS, List the names of placement algorithms. Apply Buddy System to allocate 2 MB memory space for the processes of size P1(20xN (N roll number)KB), P2(120KB), P3(512KB), P4(256K), P5(64KB), P1(128KB)? (5)

✓ Q-3) Elaborate Internal and External memory fragmentation, identify the techniques to remove Internal and External memory fragmentations? (5)

Q-4) Define PCB, pages, frames? (5)

logical or physical address scheme.

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Time: 90 minutes

Operating System (CT-353) - B

Max Marks: 20

Note: Attempt all questions.

- Q-1) Why the placement algorithms use in OS, List the names of placement algorithms.
Apply Buddy System to allocate 2 MB memory space for the processes of size P₁(20xN (N³ roll number)KB), P₂(120KB), P₃(512KB), P₄(256K), P₅(64KB), P₆(128KB)? (5)
- Q-2) Describe each state of five (5) state process models, draw figure in support answer? (5)
- Q-3) Identify the three addressing scheme of Memory, How they map with each other (5)
- Q-4) Define Boot process for an OS (5)